SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA04 **COURSE CODE:** Operating Systems

COURSE OUTCOME COVERED

CO1: *Apply the concepts of Operating System types, structures, and system calls to demonstrate their functions in various computing environments.* (BL2, BL3)

QUESTIONS: 1

Total Marks:100

Annexure A

Concept Mapping

Code of Conduct:

I Saipriya (192365065) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

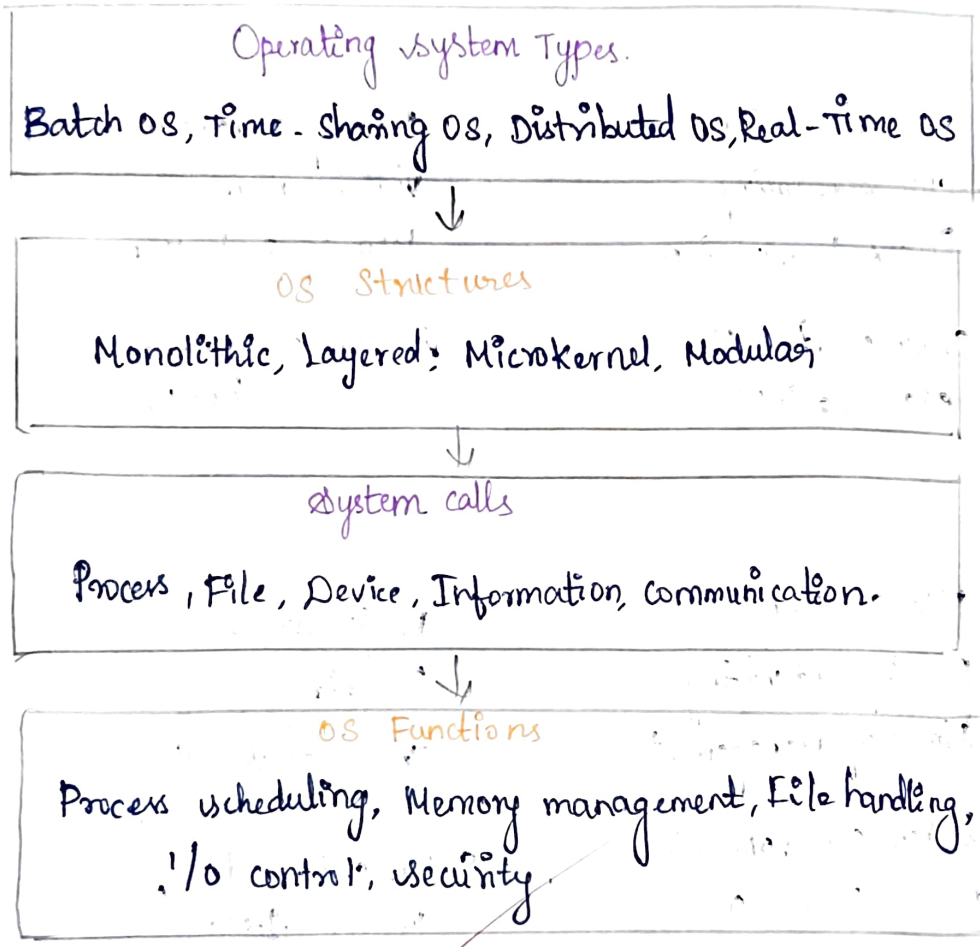
95/100

CO1 - ASSESSMENT TOOL 2

Concept Mapping

Question 1

Show a multi-level concept map integrating:
operating system Types → OS structures → System calls → OS Functions.
Your map must show cross-links.

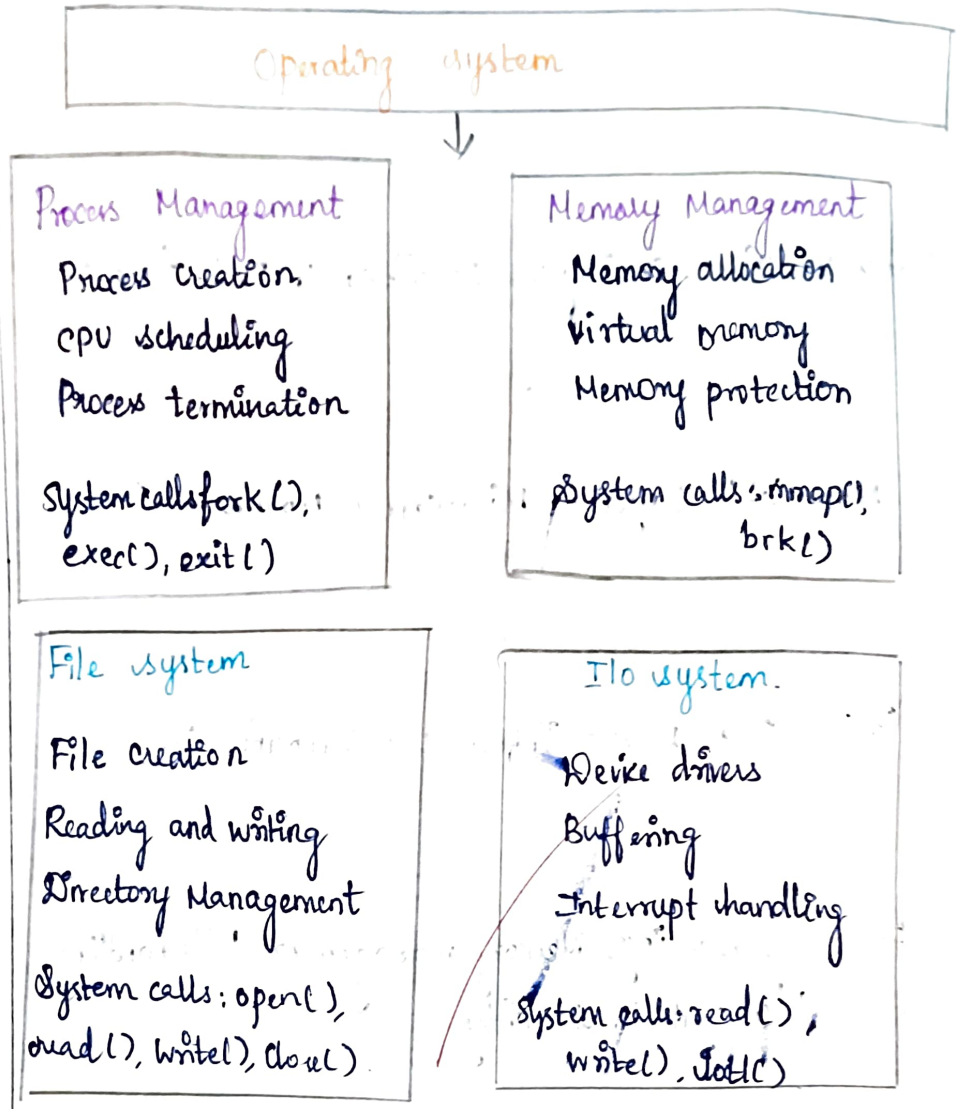


Cross-links.

→ OS type determines the service required
→ OS structure influences system call implementation.
→ System calls allow programs to request Kernel services
→ OS functions manage hardware and software resources.

Question - 2

Construct a comprehensive concept map representing: Process Management, Memory management, File systemand I/o System. Include interdependencies and indicate where system calls interact with each component.



Interdependencies.

Process Management → Memory management: processes need memory to execute.

File system → I/o system: files use storage devices for data access.

Memory Management → I/o systems: buffers support data transfer.

Process Management → I/o systems: Processes request I/o operations.

Question 3

Sketch a comparative concept map for:

Batch OS, Time-sharing OS, Distributed OS, and Real-Time OS. Highlight differences, similarities, advantages, and limitations using linking phrases.

Operating System Types

Batch OS

Working: Execute jobs in groups

Advantages: Efficient for repetitive jobs.

Limitation: Slow response

Example: Payroll processing.

Time-Sharing OS

Working: Shares CPU time among process.

Advantages: Interactive use.

Limitation: Performance may decrease under heavy.

Example: UNIX systems.

Distributed OS

Working: Coordinates multiple Computers

Advantage: Resource sharing

Limitation: Network dependency

Example: Distributed computing environments.

Real-Time OS

Working: Meets timing constraints.

Advantage: Predictable response

Limitation: Complex design.

Example: Industrial control System.

Similarities

All manage system resources and execute programs.

Differences.

Batch OS -> focuses on job processing

Time sharing OS -> focuses on user interaction

Distributed OS -> focuses on sharing resources across computers.

Real-time OS -> focuses on meeting deadlines.